

Submission DSA1-FINAL-SUB03

Question DSA1-FINAL-Q01

started with a familiar rule, but the key condition is wrong:

$7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.

I treated variables/constraints as independent, so this conclusion is not justified.

Question DSA1-FINAL-Q02

started with a familiar rule, but the key condition is wrong:

Maintain a hash set of values seen so far.

I treated variables/constraints as independent, so this conclusion is not justified.

Question DSA1-FINAL-Q03

started with a familiar rule, but the key condition is wrong:

A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$.

I treated variables/constraints as independent, so this conclusion is not justified.

Question DSA1-FINAL-Q04

started with a familiar rule, but the key condition is wrong:

Queue A; discover B,C; then dequeue B and discover D; dequeue C and discover E.

I treated variables/constraints as independent, so this conclusion is not justified.

Question DSA1-FINAL-Q05

started with a familiar rule, but the key condition is wrong:

Invariant: the prefix before index i is sorted and contains the original prefix values.

I treated variables/constraints as independent, so this conclusion is not justified.

Question DSA1-FINAL-Q06

started with a familiar rule, but the key condition is wrong:

$7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.

I treated variables/constraints as independent, so this conclusion is not justified.

Question DSA1-FINAL-Q07

started with a familiar rule, but the key condition is wrong:

Maintain a hash set of values seen so far.

I treated variables/constraints as independent, so this conclusion is not justified.

Question DSA1-FINAL-Q08

started with a familiar rule, but the key condition is wrong:

A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$.

I treated variables/constraints as independent, so this conclusion is not justified.

Question DSA1-FINAL-Q09

started with a familiar rule, but the key condition is wrong:

Queue A; discover B,C; then dequeue B and discover D; dequeue C and discover E.
I treated variables/constraints as independent, so this conclusion is not justified.

Question DSA1-FINAL-Q10

started with a familiar rule, but the key condition is wrong:

Invariant: the prefix before index i is sorted and contains the original prefix values.

I treated variables/constraints as independent, so this conclusion is not justified.